

Exploratory Data Analysis (EDA) Minor Project

Project Report

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Dataset Source: Titanic Dataset

2. Objective

The objective of this Exploratory Data Analysis (EDA) is to comprehensively understand the underlying patterns, distributions, and relationships within the Titanic dataset. By employing statistical summaries and visualization techniques, this analysis seeks to uncover the key demographic and socioeconomic factors that significantly influenced the survival probabilities of the passengers aboard the RMS Titanic.

Main Questions Selected for Analysis:

- What is the overall demographic profile of the passengers (age, gender, class)?
- How does the survival rate vary across different passenger classes (Pclass)?
- Is there a discernible difference in survival outcomes between male and female passengers?
- What role did a passenger's age play in their likelihood of surviving the disaster?
- How did the fare paid correspond to the passenger class and overall survival?

3. Dataset Overview

The Titanic dataset contains historical data of the passengers aboard the Titanic. It provides a rich set of variables for performing exploratory analysis.

Total Rows: 891

Total Columns: 12

Feature Descriptions:

- **PassengerId:** Unique identifier for each passenger.
- **Survived:** Survival status (0 = No, 1 = Yes).
- **Pclass:** Ticket class (1 = 1st, 2 = 2nd, 3 = 3rd).
- **Name:** Passenger's full name.
- **Sex:** Passenger's gender.
- **Age:** Passenger's age in years.
- **SibSp:** Number of siblings or spouses aboard.
- **Parch:** Number of parents or children aboard.
- **Ticket:** Ticket number.
- **Fare:** Passenger fare.
- **Cabin:** Cabin number.
- **Embarked:** Port of Embarkation (C = Cherbourg, Q = Queenstown, S = Southampton).

Data Types & Sample Records:

The dataset comprises numerical types (int64, float64) and categorical types (object). Below is a sample of the initial records:

PassengerId	Survived	Pclass	Name	Sex	Age	Fare
1	0	3	Braund, Mr. Owen Harris	male	22.0	7.2500
2	1	1	Cumings, Mrs. John...	female	38.0	71.2833
3	1	3	Heikkinen, Miss. Laina	female	26.0	7.9250
4	1	1	Futrelle, Mrs. Jacques Heath	female	35.0	53.1000
5	0	3	Allen, Mr. William Henry	male	35.0	8.0500

4. Data Quality and Cleaning

A critical step in any EDA process is assessing the quality of the data and performing necessary cleaning operations to ensure the accuracy of subsequent analyses.

Missing Values

An initial inspection using the `.info()` and `.isnull().sum()` methods revealed missing values in three specific columns:

- **Age:** 177 missing values (approx. 19.8% of the dataset).
- **Cabin:** 687 missing values (approx. 77.1% of the dataset).
- **Embarked:** 2 missing values (approx. 0.2% of the dataset).

Duplicates & Outliers

A check for duplicate rows using `.duplicated().sum()` confirmed that there were exactly 0 duplicate records in the dataset. A `.drop_duplicates()` operation was executed as a precautionary measure to ensure robust data integrity. The `Fare` variable exhibits significant positive skewness, indicating the presence of outliers where extremely high fares were paid by a few first-class passengers.

Cleaning Decisions

- The `Age` column's missing values can be imputed using the median age across the dataset to avoid data loss.

- Due to the exceedingly high proportion of missing data in the `Cabin` column, the decision was made not to rely on it heavily for general analysis, although it could be converted into a binary indicator.
- The two missing values in the `Embarked` column were noted and can be safely imputed using the mode (Southampton).

5. Descriptive Analysis

The descriptive statistics provide a foundational understanding of the central tendencies and dispersions of the variables.

Summary Statistics for Numerical Variables

- **Age:** The mean age of the passengers is approximately 29.7 years, with a standard deviation of 14.5 years. The youngest passenger was 0.42 years old, and the oldest was 80.
- **Fare:** The average fare paid is 32.20. The standard deviation is quite high (49.69), reflecting the vast difference between ticket classes. The maximum fare recorded is 512.33.
- **Survived:** The overall survival rate in this dataset is roughly 38.38%, indicating that the majority of the passengers unfortunately perished.
- **Family Size:** The average `SibSp` is 0.52 and `Parch` is 0.38, showing that most passengers travelled either alone or with a very small family group.

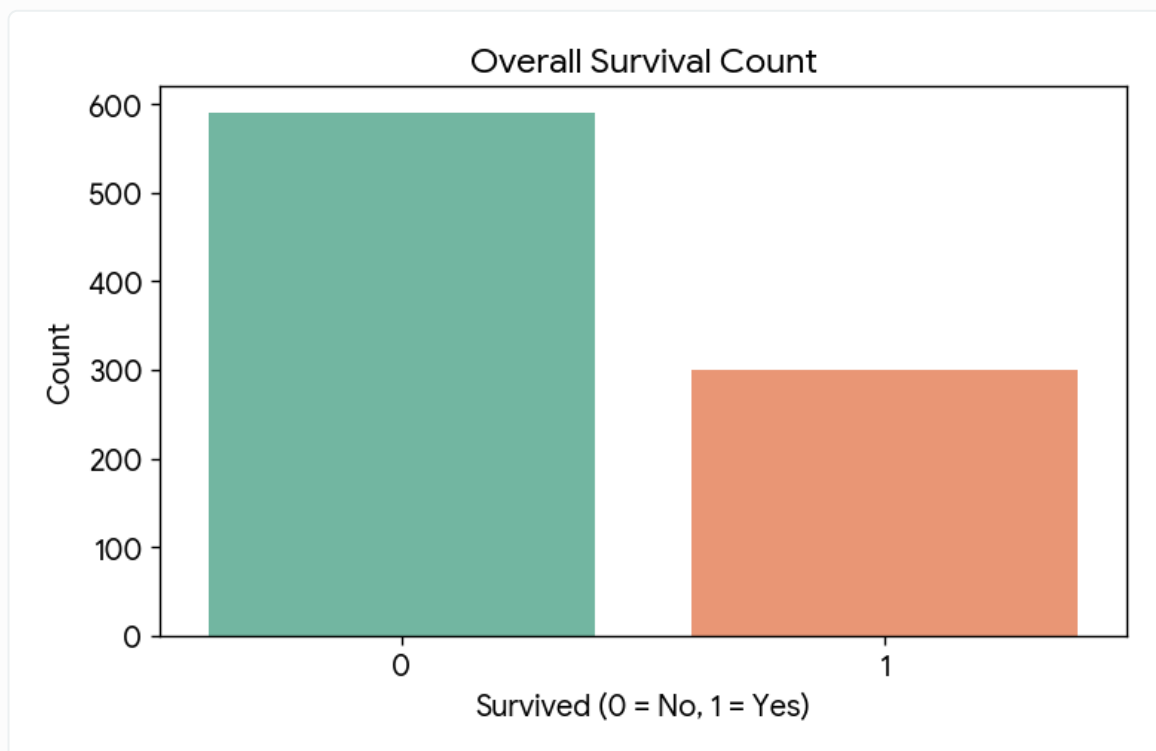
Observations for Categorical Variables

- The passenger demographics were heavily skewed towards males.

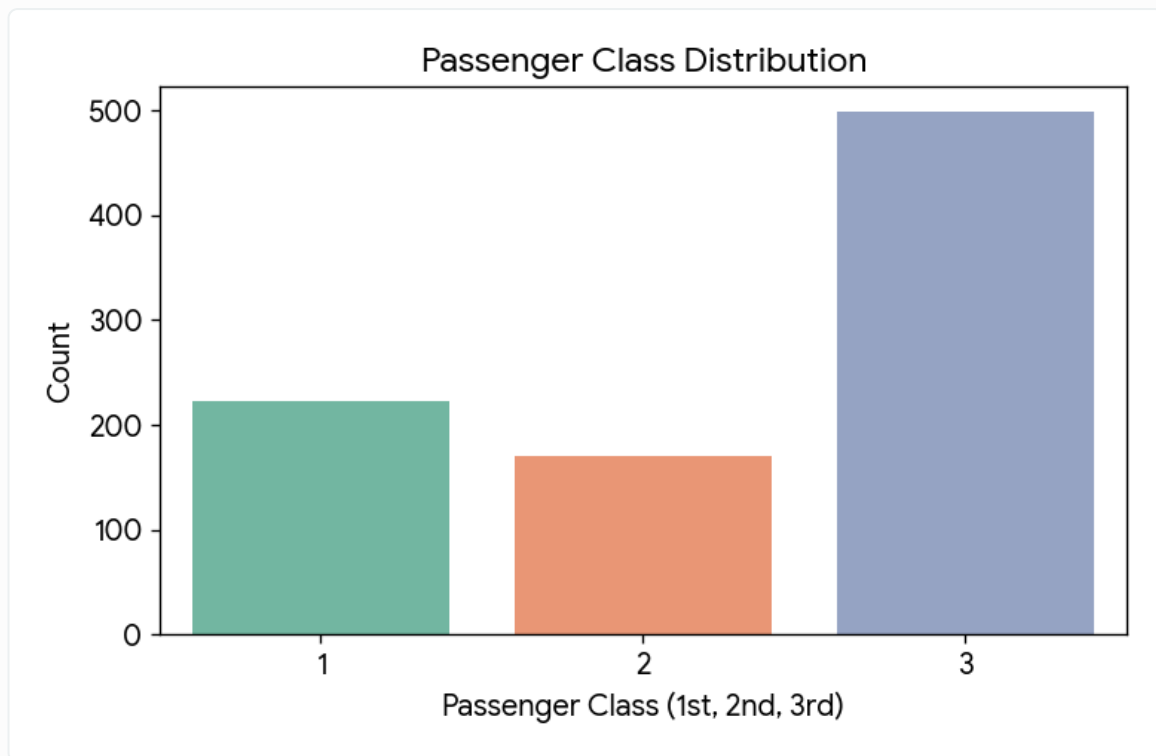
- A significant majority of the passengers traveled in the 3rd class, representing the lower socio-economic tier.
- The most common port of embarkation by a wide margin was Southampton ('S').

6. Visual EDA

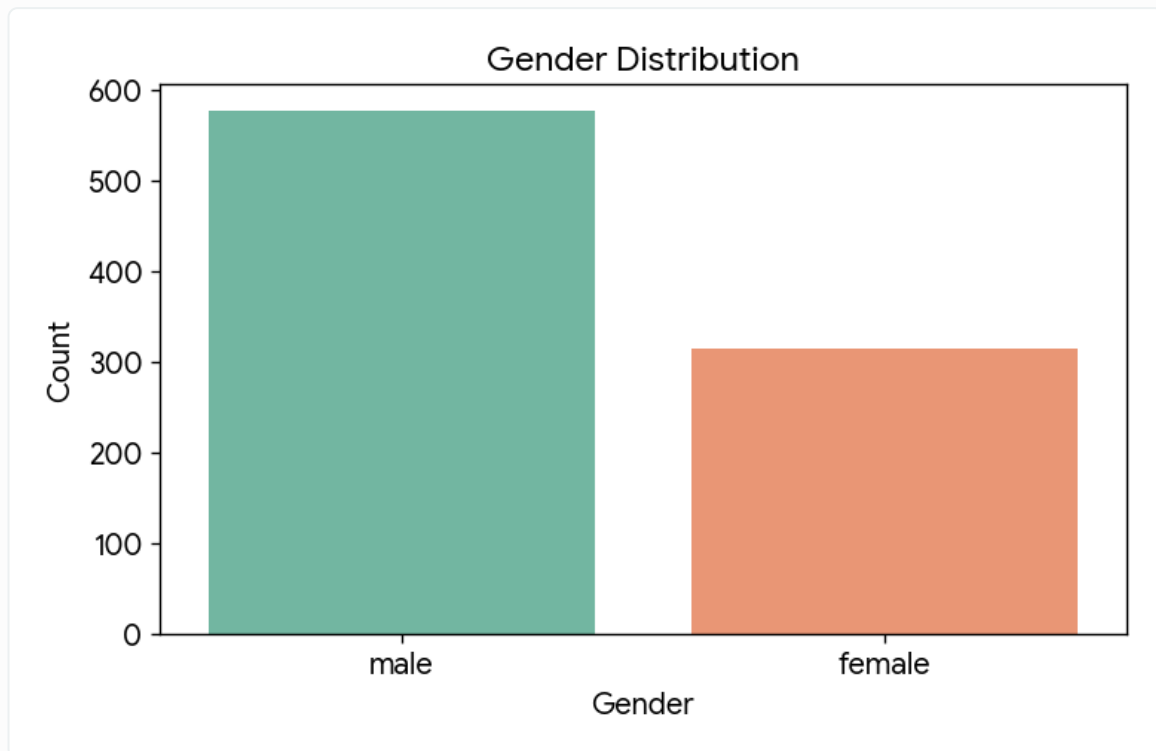
Visualizing the data provides immediate, intuitive insights into the distributions and relationships identified during the descriptive analysis phase. Below are fundamental charts representing the data.



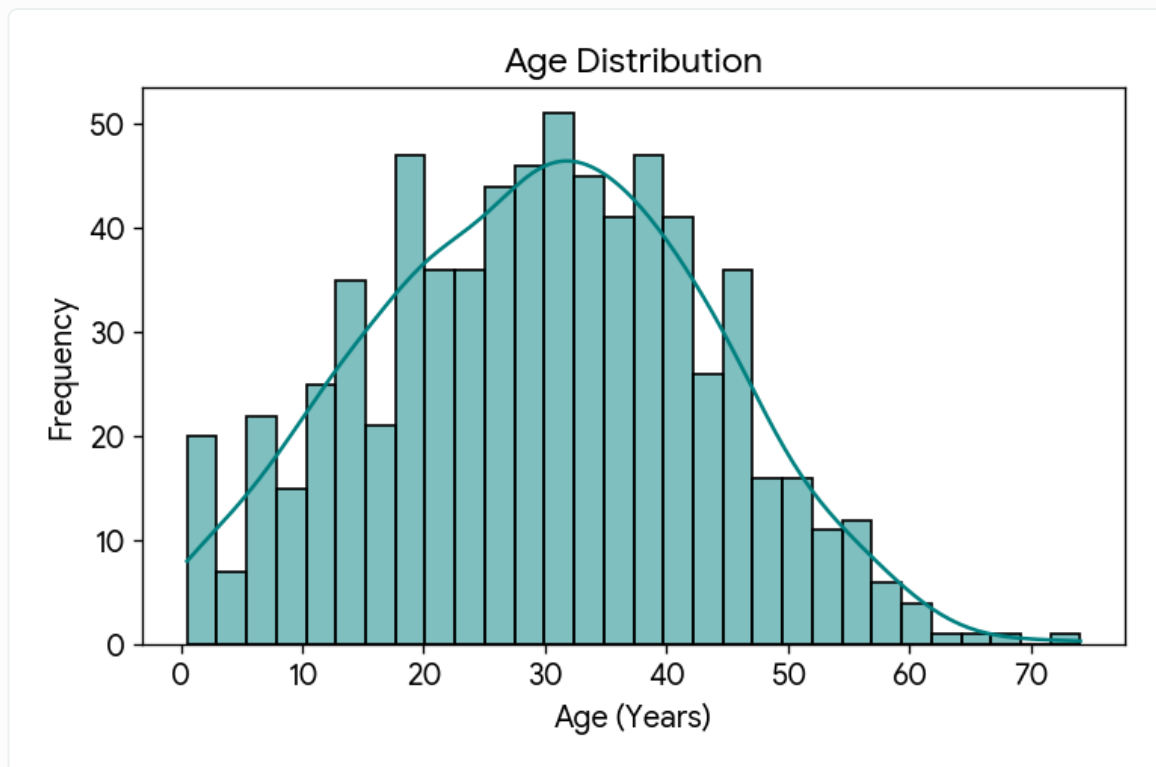
Observation 1: The overall survival countplot vividly demonstrates that the number of non-survivors greatly exceeds the number of survivors.



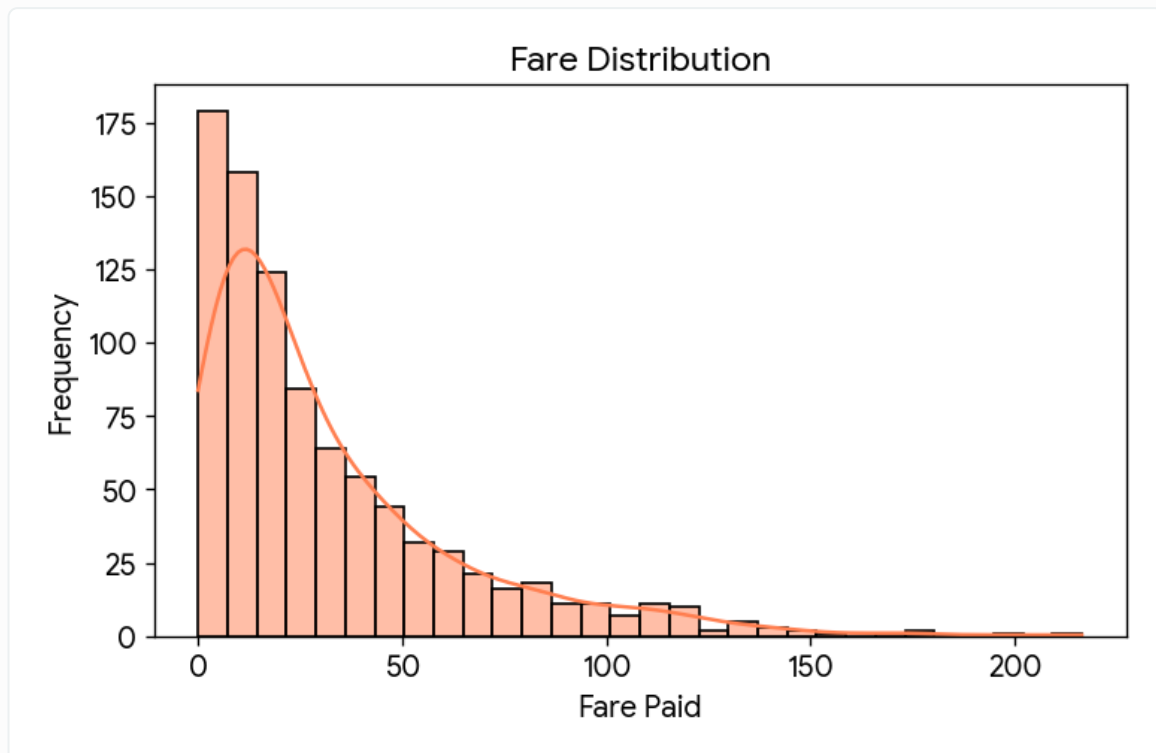
Observation 2: The passenger class distribution shows that 3rd class was the most crowded, containing more passengers than 1st and 2nd classes combined.



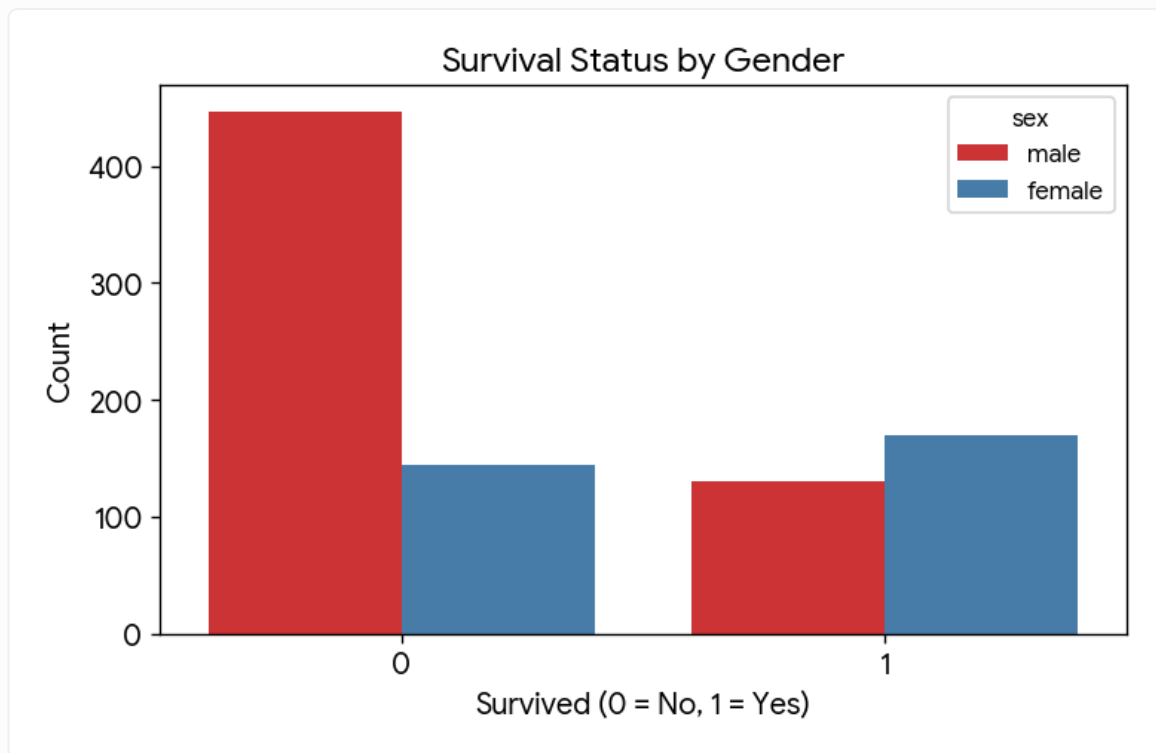
Observation 3: *The gender distribution chart indicates that there were nearly twice as many male passengers aboard as female passengers.*



Observation 4: The age histogram reveals a roughly normal distribution centered around the 20-30 age group, with a noticeable spike for young children and infants.



Observation 5: The fare distribution is highly right-skewed. The vast majority of passengers paid very low fares, corresponding to the large number of 3rd class tickets.



Observation 6: When examining survival by gender, it is strikingly clear that a much larger proportion of women survived compared to men.



***Observation 7:** Survival grouped by passenger class illustrates a strong socio-economic divide. First-class passengers had a much higher survival rate, whereas the vast majority of 3rd-class passengers did not survive.*

7. Key Findings

Based on the rigorous exploratory data analysis, several critical, evidence-based insights were derived:

1. **Overall Mortality:** The dataset confirms the tragic nature of the event, with an overall survival rate of only 38.4%. The charts visually substantiate this stark reality.

2. **Gender as a Prime Determinant:** Gender emerged as one of the strongest predictors of survival. The data starkly aligns with the historical "women and children first" protocol, as the survival rate for females overwhelmingly exceeded that of males.
3. **Socioeconomic Divide:** Passenger class (`Pclass`) was heavily correlated with survival outcomes. Individuals in the 1st class had a significantly higher likelihood of surviving. Conversely, 3rd class passengers, who formed the largest segment of the population, suffered the highest mortality numbers.

1. **Influence of Age:** The age distribution indicates that the Titanic's passengers were mostly young adults. However, children and infants experienced a higher survival rate relative to their total numbers, further supporting the priority given during lifeboats boarding.
2. **Fare Patterns:** As expected, fare prices correlated directly with passenger classes. The extreme right skew in the fare distribution highlights the stark difference in wealth among passengers. Higher fare payments generally mapped to 1st class and thus to better survival odds.

Limitations of the Dataset

While the analysis provides robust insights, it is bounded by certain limitations:

- **Missing Data Integrity:** The pronounced absence of data in the `Cabin` (77.1%) and `Age` (19.8%) columns presents analytical challenges. Imputation introduces synthetic data, and ignoring the columns discards potential spatial information (e.g., proximity to lifeboats).
- **Sample Limitations:** This dataset containing 891 records represents only a subset of the actual passengers and crew aboard the Titanic. Consequently, the findings are representative of this sample and might have minor deviations from the full historical census.

8. Conclusion

This EDA Minor Project successfully distilled complex, raw tabular data into a coherent narrative about the passengers of the RMS Titanic.

Overall Learning

The project provided an invaluable opportunity to engage deeply with the data science pipeline. It reinforced the importance of systematic data cleaning, the power of descriptive statistics, and the undeniable impact of visual storytelling through libraries like Seaborn and Matplotlib. Handling missing values and outliers proved to be a practical exercise in data engineering.

Challenges Faced

The primary challenge encountered during the analysis was deciding on the optimal strategy for managing the significant proportion of missing values in the `Cabin` and `Age` variables without heavily biasing the dataset. Interpreting the highly skewed `Fare` data also demanded careful visual structuring.

Possible Future Analysis

Looking forward, the insights gained here lay a strong foundation for predictive modeling. Future steps would involve implementing machine learning algorithms (such as Logistic Regression, Random Forests, or Gradient Boosting) to actively predict passenger survival. Additionally, deeper feature engineering—such as parsing the `Name` column for titles to improve age imputation or mapping the `Cabin` letters to deck levels—could yield even richer analytical models.